

Fuel Inventory Optimization Analysis

Course Project Report

Python for Data Analysis (BU.510.615)

Group 8

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1 Introduction

The challenge of fuel procurement for a gas station network lies in balancing conflicting objectives: capturing volume discounts while minimizing inventory costs and maintaining service levels. Our network, comprised of eight gas stations and 23 fuel tanks, faced significant procurement inefficiencies. Current practices emphasized frequent, small orders to minimize inventory, but systematically failed to capture supplier discounts. Preliminary analysis revealed that despite tiered discounts ranging from 0 cents to 4 cents per liter, 65.4% of orders were not receiving discounts.

This project analyzed fuel procurement practices using five primary datasets, covering 954 days from January 2017 to August 2019: locations and tank specifications for eight gas stations and 23 fuel tanks, 2,869 fuel purchase, time-series fuel level measurement data. Our objectives were to quantify current inefficiencies, design an optimized reordering strategy to maximize discounts while maintaining safety stock, assess system-wide risks and cost-saving potential, and develop a phased implementation roadmap.

2 Data Processing

2.1 Data Loading and Consolidation

Our analysis begins with five primary datasets. We standardized all column names to UPPER CASE with underscores for consistency. The two fuel level files required special handling. We merged these files using a multi-step process.

First, we aligned column structures between the two files. Second, we identified and removed duplicate records. These duplicates accounted for 3.2% of total records. Third, we converted all timestamps to `datetime64[ns]` format for proper time series analysis. Finally, we dropped records with null values to ensure data quality.

Figure 1 confirms successful datetime parsing. All timestamps now share a consistent format.

```
TANK_ID      object
FUEL_LEVEL   float64
TIMESTAMP    datetime64[ns]
dtype: object
```

Figure 1: Data types after timestamp conversion showing successful datetime parsing

2.2 Data Integration

We have designed a star database structure for convenient and efficient analysis. This mode consists of one dimension table and two fact tables.

The dimension table combines storage tank information and location data. We use site identifiers to connect the tank dataset to the location dataset. This comprehensive view contains all 23 tanks distributed across 8 sites. Each record includes geographical coordinates, the name of the gas station, fuel tank capacity and fuel type.

Figure 2 displays the complete dimension table structure. Each row provides comprehensive information about tank location and specifications. This structure allows us to analyze patterns at both the individual tank level and the station level.

```
<class 'pandas.core.frame.DataFrame'>
Int64Index: 23 entries, 0 to 22
Data columns (total 9 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   TANK_ID                               23 non-null     object
1   TANK_LOCATION                         23 non-null     int64
2   TANK_NUMBER                           23 non-null     int64
3   TANK_TYPE                             23 non-null     object
4   TANK_CAPACITY                         23 non-null     int64
5   GAS_STATION_NAME                     23 non-null     object
6   GAS_STATION_ADDRESS                  23 non-null     object
7   GAS_STATION_LATITUDE                 23 non-null     float64
8   GAS_STATION_LONGITUDE                23 non-null     float64
dtypes: float64(2), int64(3), object(4)
memory usage: 1.8+ KB
```

Figure 2: Integrated station dimension table with location and capacity information

The first fact table tracks the changes in fuel inventory levels over time. This time series data captures the dynamic characteristics of fuel consumption and replenishment. The second fact table records the purchase transaction. Each invoice records the date, quantity, cost and fuel type of each purchasing activity.

2.3 Data Quality Assessment

We employed the `missingno` library to systematically identify data quality issues. Our analysis revealed 38 invoice records containing missing values. This represents 1.3% of the total invoice dataset.

Figure 3 visualizes the missing value pattern. The heat map shows that missing values only occur in complete rows. No partial records exist where only some fields are missing. This mode strongly indicates a data input error rather than a system collection issue or structural problem of the data source.

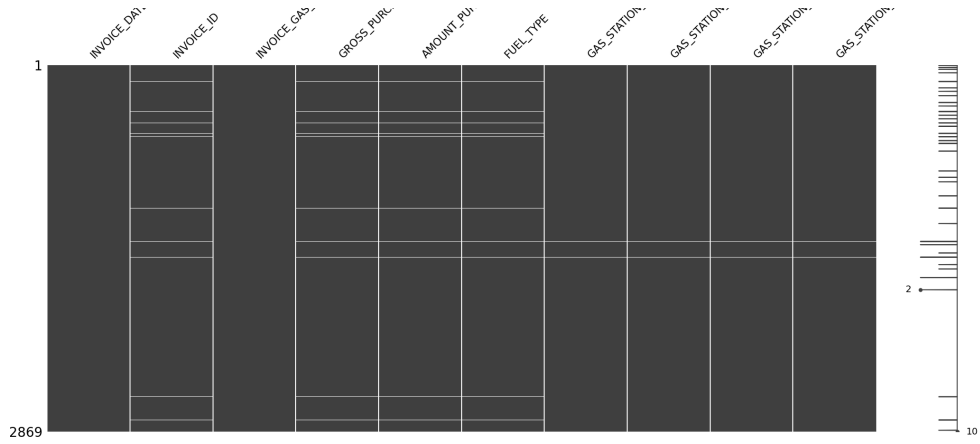


Figure 3: Missing value patterns showing complete rows with missing data

Based on this assessment, we removed all incomplete records from our analysis. The cleaned dataset contains **2,831 valid invoices**, representing 98.7% of the original data.

3 Exploratory Analysis

Having established a clean dataset, we now examine the characteristics of our gas station network.

3.1 Station Infrastructure Analysis

3.1.1 Tank Distribution Across Stations

The physical infrastructure varies considerably across our eight stations. Figure 4 quantifies this variation by showing tank counts per location.

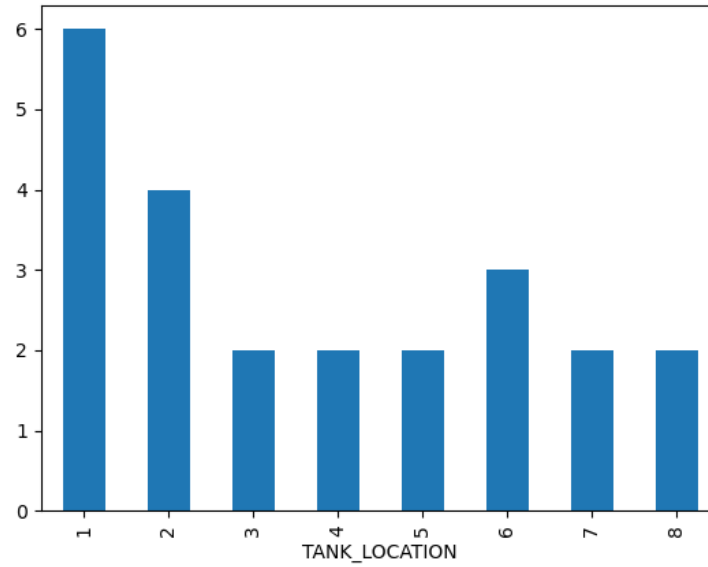


Figure 4: Tank count by station showing Station 1 as the largest facility

There are six oil tanks at Station 1. This makes it the largest facility in our network. There were four tanks following at the second stop. Each of the remaining supply stations has 2 to 3 storage tanks. Larger stations have greater flexibility for consolidated bulk orders. They can coordinate the purchase of multiple tanks to achieve a higher discount level.

3.1.2 Fuel Type Distribution

Different fuel types serve different market segments. Figure 5 breaks down total capacity by fuel type.

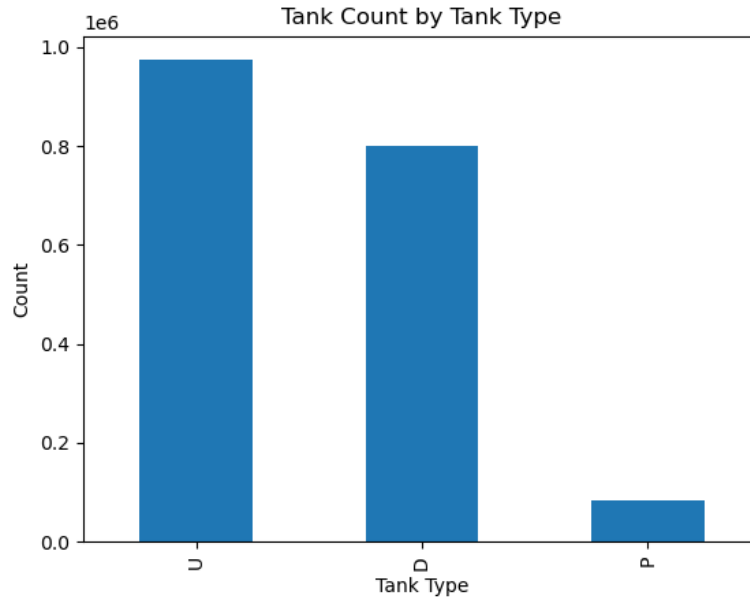


Figure 5: Fuel type distribution (D=Diesel, U=Regular, P=Premium)

Diesel fuel accounts for 52% of total capacity. This is equivalent to 970,000 liters of all diesel fuel tanks. Regular gasoline represents 43% of capacity with 800,000 liters total. Premium fuel serves a niche market with only 5% of capacity (80,000 liters).

This distribution reveals where the optimization work will generate the greatest return. Due to the high output of diesel and common fuels, they offer the best opportunities.

3.1.3 Tank Capacity Analysis

Individual tank capacity directly constrains maximum order sizes. Larger tanks enable access to deeper discount tiers. Figure 6 shows average tank capacity by station.

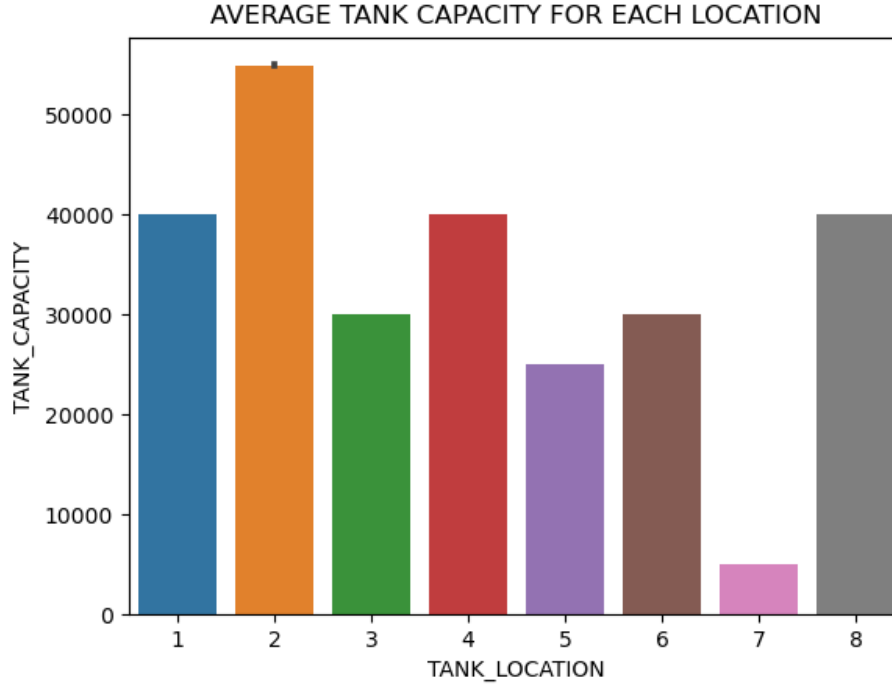


Figure 6: Average tank capacity by station

It can be seen from the figure that Station 2 is in the lead, with an average capacity of 55,000 liters. At this size, when fully filled, the fuel tank can enjoy a maximum discount of 4 cents per liter. The average capacity of Station 1 is 40,000 liters. This way, you can continuously get a discount of 3 cents or 4 cents. Station 7 presents a stark contrast with only 5,000L average capacity.

Capacity differences create a tiered opportunity structure across stations. Station 7 requires urgent expansion from 5,000L to 25,000L to unlock 3¢ discounts, as its current capacity limits the discount to a maximum of 0 to 2 cents. Station 5 is worth considering expanding from 10,000 liters to 25,000 liters to obtain a more consistent 3-cent discount. Station 1, Station 2, Station 4 and Station 8 have sufficient capacity for optimal bulk purchasing without infrastructure investment.

3.2 Procurement Pattern Analysis

In addition, we also examined the seasonal trends and order frequency patterns to understand the fuel purchase situation.

3.2.1 Seasonal Demand Variation

Fuel demand exhibits strong seasonal patterns. These patterns affect both costs and volumes. Figures 7 and 8 illustrate this seasonality clearly.

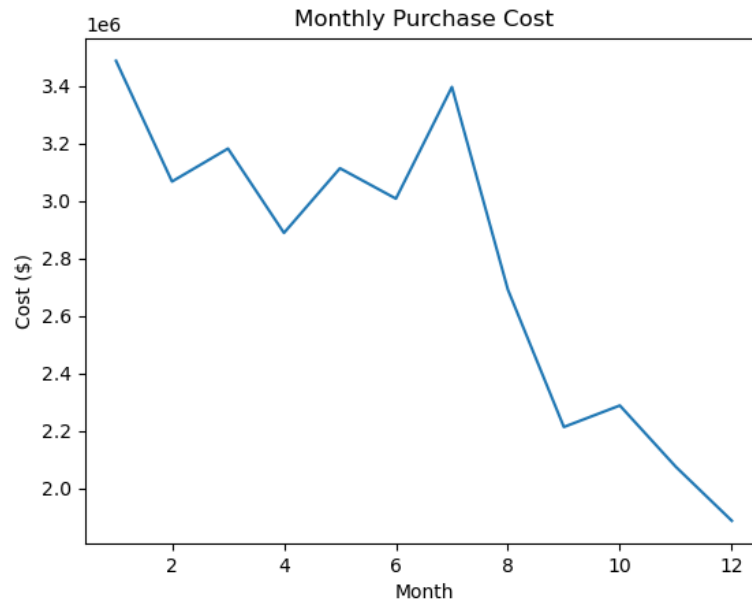


Figure 7: Monthly fuel costs showing seasonal variation

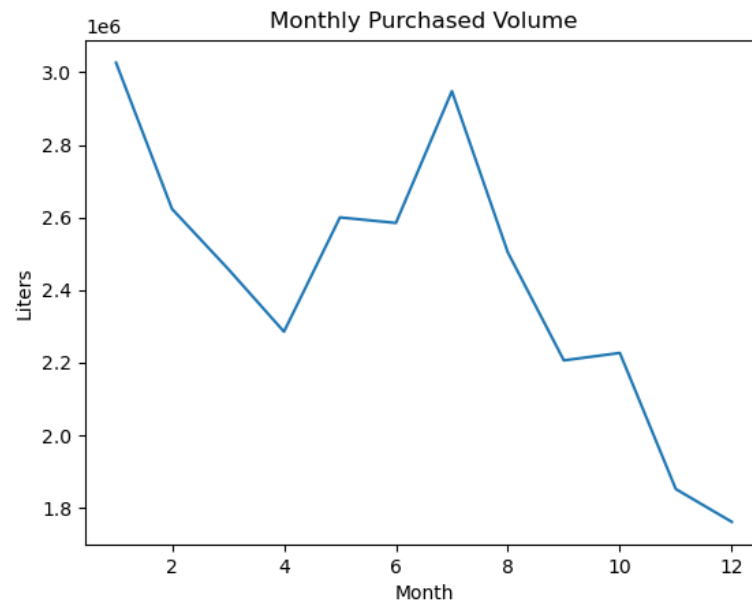


Figure 8: Monthly volume mirroring cost patterns

Month 7 (summer) represents the annual peak with costs of \$3.45 million and volume of 3.0 million liters, while Month 12 (winter) marks the trough at \$1.88 million and 1.75 million liters, representing a 45% decline. This obvious seasonal variation requires a flexible reordering policy to respond to actual inventory levels rather than a fixed schedule, as static policies

may lead to shortages during periods of high demand or overstocking during periods of low demand.

3.2.2 Order Frequency Patterns

Figure 9 provides a longitudinal view of procurement behavior. The analysis spans the complete 954-day study period from January 2017 to August 2019.

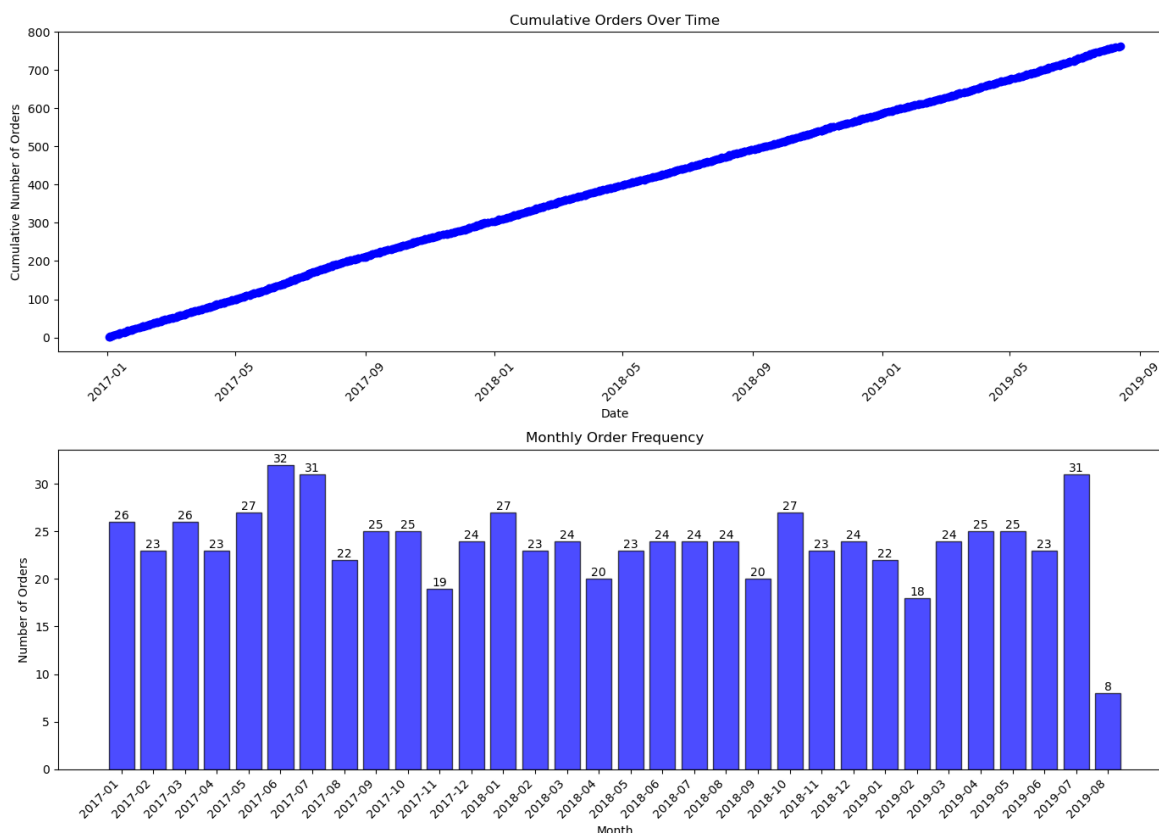


Figure 9: Order timeline showing cumulative growth and monthly frequency distribution

The cumulative line shows a nearly linear growth, indicating that despite monthly fluctuations, overall demand remains consistent. The monthly bar chart shows the seasonal peaks in July 2017 (32 orders) and July 2018 (31 orders), while the winter troughs in December 2017 (19 orders) and August 2019 (8 orders). The system processed a total of 762 orders within 954 days, averaging 23.8 orders per month, which is close to one order per day. Such a high frequency indicates that procurement is a reactive rather than a strategic approach. Each order generates administrative overhead including processing requisitions, receiving deliveries, inspecting shipments, and processing payments. An optimized bulk purchasing strategy could reduce order frequency by over 90%, freeing procurement staff to focus on strategic rather than tactical activities. These improvements in operational efficiency complement the direct cost savings brought about by discount capture.

4 Inflation Impact Analysis

Understanding true cost requires accounting for inflation. We incorporated Statistics Canada's monthly Consumer Price Index data to reveal real costs.

4.1 Inflation Rate Data and Methodology

We collected official monthly inflation rates for 2017 from Statistics Canada. These year-over-year rates capture how prices evolved throughout our study period. The table below summarizes these rates.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov
Dec											
Rate (%)	2.1	2.0	1.9	1.6	1.4	1.0	1.0	1.4	1.6	1.9	2.1
1.9											

The average annual inflation rate was 1.66%. Rates ranged from a low of 1.0% in mid-year to highs of 2.1% in January and November. We applied these rates to adjust all nominal prices to real (inflation-adjusted) values.

4.2 Four-Dimensional Inflation Analysis

Figure 10 presents inflation's impact from four complementary perspectives. Each panel reveals different aspects of how inflation eroded purchasing power during 2017.

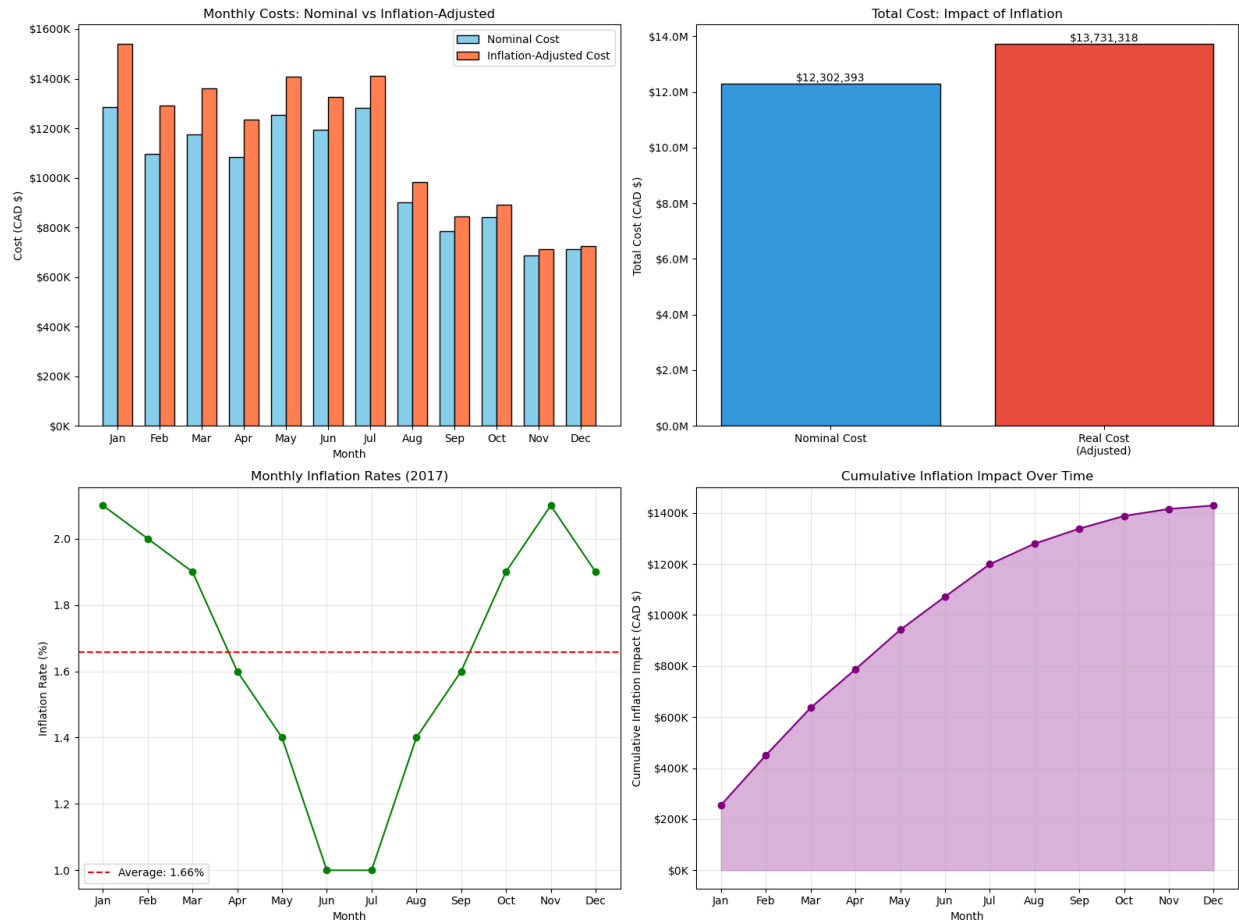


Figure 10: Four-panel inflation analysis showing monthly costs, total burden, rate dynamics, and cumulative impact

The top-left panel compares monthly nominal versus real costs. Early-year purchases (January to April) show the largest gaps with inflation rates of 1.6% to 2.1%, while summer months (June to July) show smaller discrepancies at 1.0% inflation. Fall rebound reopened gaps as inflation returned to 1.9% to 2.1%.

The top-right panel quantifies total burden. Nominal costs totaled \$12,302,393 versus real costs of \$13,731,318, representing a hidden purchasing power loss of **\$1,428,925** (11.6% penalty).

The bottom-left panel tracks inflation rate dynamics showing a U-shaped pattern typical of commodity-linked inflation. The year began at 2.1%, dropped to 1.0% mid-year, and rebounded to 1.9% to 2.1% by year-end.

The bottom-right panel shows cumulative inflation impact reaching **\$1.43 million** by December. Growth was steepest in Q1 and Q4 when inflation ran highest, with a plateau during summer's low-inflation period.

4.3 Strategic Implications for Procurement

Inflation affects procurement optimization in three aspects: it reveals that actual savings exceed nominal calculations, creates time value in procurement time, and increases the value of the discount layer over time. In our case study, the nominal bulk order saved \$10,370,989, but inflation adjustment raises real savings to **\$11,798,914**, a 13.8% increase reflecting purchasing power preservation.

Due to the rising inflation rate, the actual cost of earlier purchases was higher, which strengthened the justification for immediate implementation, as each monthly delay would lead to the abandonment of discounted savings and the risk of inflation. Additionally, discount tiers gain value over time: a 3¢/L discount earned in January equals 3.06¢ by December due to dollar erosion. The table below compares nominal versus real metrics using inflation-adjusted figures.

Metric	Nominal	Real
Current cost	\$12,302,393	\$13,731,318
Optimized cost	\$1,931,404	\$2,149,570
Savings	\$10,370,989	\$11,581,748
Rate	84.30%	84.35%
Inflation impact	—	+\$1,210,759

From the table we find that real annual savings exceed \$11.6 million, surpassing nominal figures by \$1.2 million. Implementation delay costs approximately \$864,000 monthly in direct discount savings plus additional inflation erosion, creating a total opportunity cost exceeding one million dollars per month.

5 Case Study: Tank 10 Optimization

We select Tank No. 10 from the first station as the main example for the detailed case study. This tank exhibits characteristics typical of our network.

5.1 Tank Selection Rationale

Tank 10 has several attributes that make it representative. The fuel tank capacity is 40,000 liters, which is a common capacity in our network. It stores regular gasoline, which is the second largest type of fuel after diesel. It operates at Station 1, which is our largest facility and has six storage tanks. Most importantly, it demonstrates a consistent demand pattern that conforms to policy optimization.

5.2 Current Performance Analysis

5.2.1 Order Size Distribution Problems

Current procurement practices reveal a fundamental flaw. Figure 11 exposes this issue through order size distribution analysis.

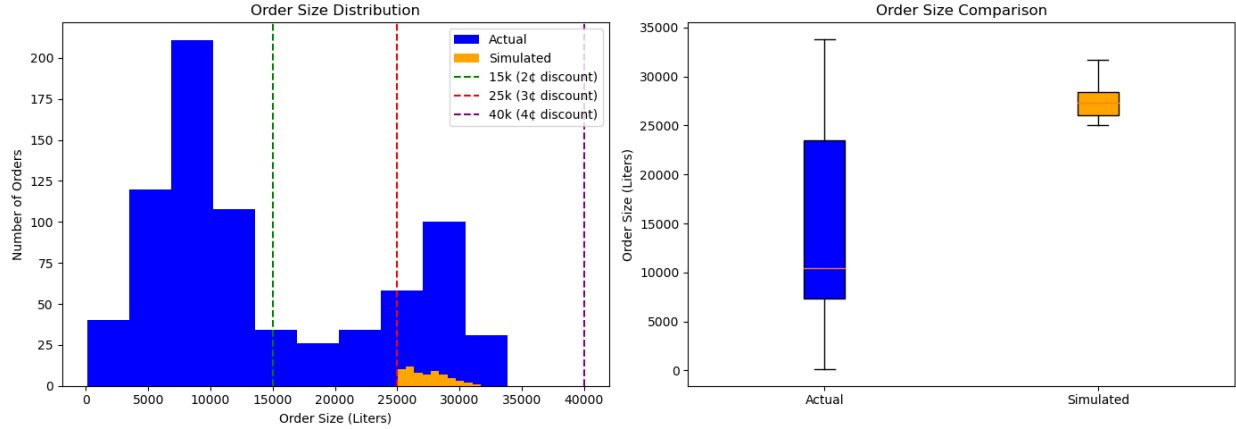


Figure 11: Current vs. Optimized Order Sizes

From the figure, average order size measures just 14,260 liters, falling below the 15,000L threshold required for any discount. The histogram shows 65.4% of orders receive zero discount while only 23.2% capture the 3¢ discount across 762 total orders. The root cause lies in order triggering logic that initiates purchases too frequently while inventory remains too high, producing small quantities that consistently miss discount tiers. This represents a classic "penny-wise, pound-foolish" approach optimizing for inventory minimization rather than total cost minimization.

5.2.2 Inventory Evolution Analysis

Figure 12 traces actual inventory levels throughout the study period. The graph reveals frequent small refills that maintain inventory between 20,000L and 35,000L. The tank rarely approaches empty state.

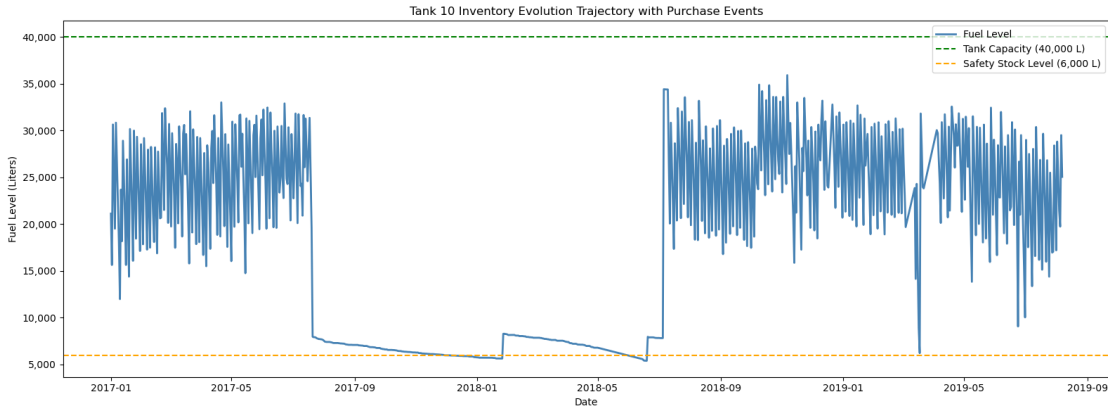


Figure 12: Tank 10 inventory over time showing frequent small refills

The figure shows that the average daily consumption is 845 liters, with significant consistency. This stable consumption rate simplifies optimization. We can predict inventory depletion

with high confidence. Current policy maintains excessive safety stock. The tank rarely drops below 50% capacity. This conservatism prevents the optimal order size.

5.3 Optimized Policy Design

5.3.1 Policy Parameters

We designed a reorder point policy that refills the tank to full capacity:

Reorder Point: 15,000L

- Triggers order when inventory drops to 15,000L
- Provides 17.7 days safety stock ($15,000\text{L} \div 845 \text{ L/day}$)
- Far exceeds industry standard of 3–7 days

Refill Strategy: Fill to capacity (40,000L)

- Order quantity = 40,000L - current inventory
- When triggered at 15,000L → order 25,000L
- 25,000L captures 3¢/L discount (25,000–40,000L tier)

This simple policy ensures every order qualifies for the 3¢ discount while maintaining conservative safety margins.

5.3.2 Simulation Results

We simulated one year of operations under the optimized policy. Figure 13 compares results against actual historical performance.

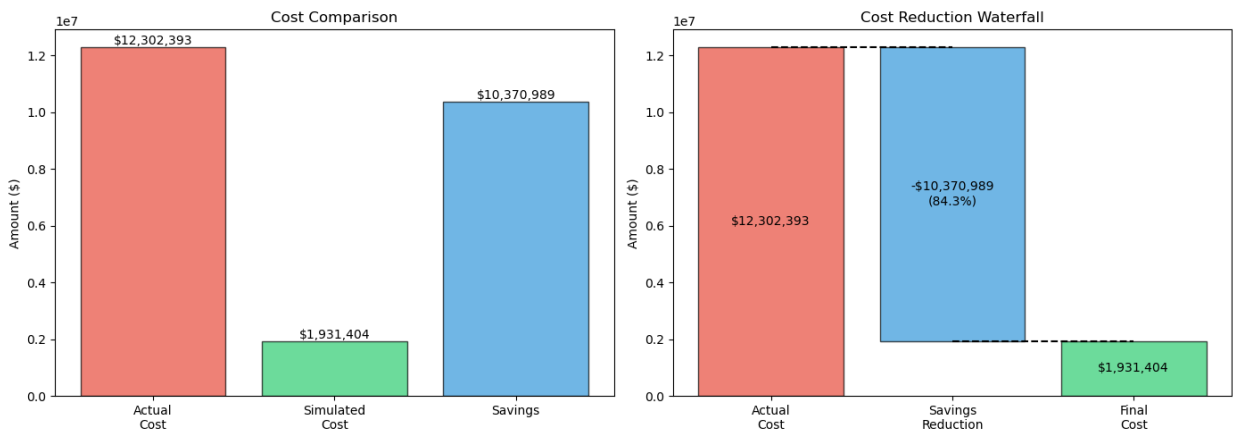


Figure 13: Cost comparison showing 84% savings with optimized strategy

The bar chart dramatically illustrates the opportunity. Current annual costs reached \$12,302,393. Optimized costs drop to \$1,931,404. This represents **\$10,370,989 in annual savings**, an 84.3% reduction. The following table details the transformation across multiple dimensions.

Metric	Current	Optimized
Total Cost	\$12,302,393	\$1,931,404
Savings	—	\$10,370,989 (84.3%)
Average Order Size	14,260L	25,000L
Orders per Year	762	64
Zero Discount Rate	65.4%	0%
3¢ Discount Rate	23.2%	100%

Order frequency plummets from 762 to 64 annually, representing a 91.6% reduction in transaction volume. Every order now captures the 3¢ discount while zero-discount purchases become extinct under the new policy.

5.4 Comprehensive Performance Evaluation

Figure 14 provides three complementary perspectives on the optimization impact.

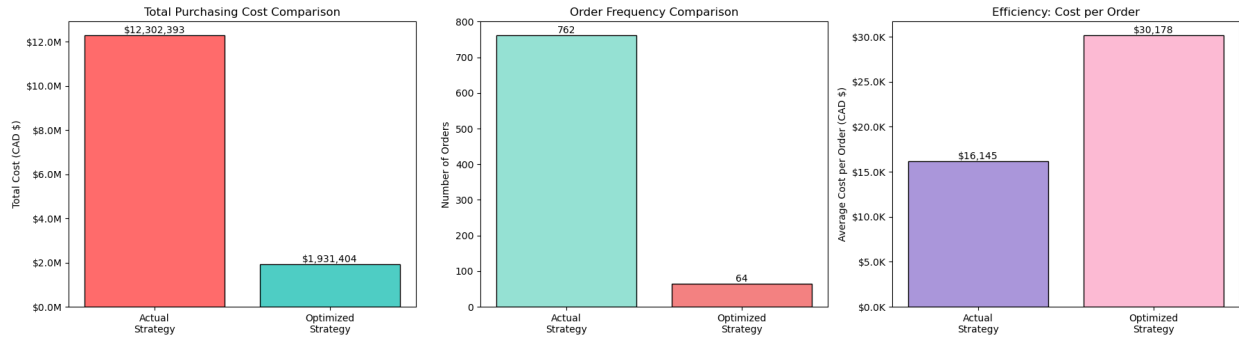


Figure 14: Three-metric comparison: total cost, order frequency, and cost per order

The left panel shows total cost efficiency: current strategy costs \$12,302,393 annually versus optimized strategy at \$1,931,404, yielding \$10,370,989 savings (84.30% reduction).

The center panel reveals operational efficiency gains with orders dropping from 762 annually (2.1 per day) to 64 (5.3 days between orders), eliminating 698 transactions (91.60% decline). The right panel examines transaction economics: while cost per order rises from \$16,145 to \$30,178 (87% increase), this is intentional as larger orders capture deeper discounts per liter, with the 91.6% order frequency reduction more than compensating for higher individual transaction costs.

Risk Assessment: The 15,000L reorder point provides 17.7 days of safety stock, exceeding industry standards of 3 to 7 days by a factor of 2.5 to 6. This conservative design buffers against demand spikes up to 50% and supplier delivery delays up to two weeks, maintaining stockout risk near zero under normal operating conditions.

6 System-Wide Risk and Opportunity Analysis

After verifying the optimization potential of Tank No. 10, we now study a broader network. Understanding the risk distribution within the system can guide the rollout of priorities.

6.1 Comprehensive Inventory Risk Assessment

Figure 15 presents three complementary views of inventory risk across all 23 tanks. Together, these perspectives identify critical vulnerabilities and opportunities.

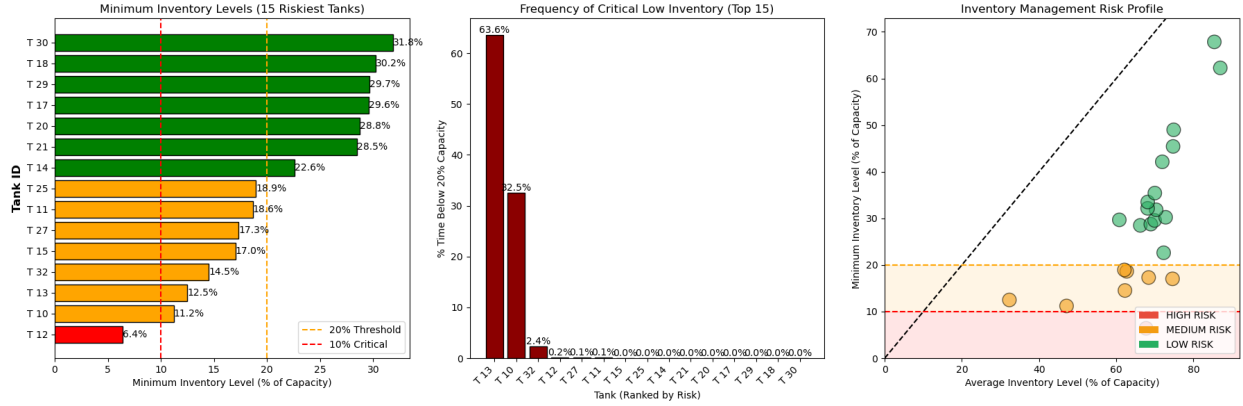


Figure 15: System-wide inventory risk analysis with three perspectives

We classify tanks into three risk categories based on minimum inventory levels. High-risk (red) tanks maintain minimum inventory below 10% of capacity. These tanks face frequent stockout danger. Medium-risk (orange) tanks hold minimum inventory between 10% and 20% capacity. Low-risk (green) tanks maintain minimum inventory above 20% capacity.

6.1.1 Critical Findings and Implementation Priority

We find that Tank 12 emerges as the most urgent concern with minimum inventory at only 6.4% of capacity, spending 63.6% of its time in the critical zone. A high-risk cluster of 8 tanks (T12, T10, T13, T32, T15, T27, T11, T25) all fall below the 20% minimum inventory threshold, representing 35% of system capacity. Conversely, 10 tanks demonstrate effective management with average inventory above 30% and minimums above 20%, serving as benchmarks for best practices.

The risk assessment provided information for our phased rollout: in Phase 1a, emergency intervention was carried out for Tank 12; in phase 1b, the remaining 7 high-risk tanks were adjusted through reordering points; in phase 2, medium-risk tanks were optimized; and in phase 3, the bulk purchase efficiency of low-risk tanks was maximized. Risk level determines reorder point settings: high-risk tanks at 25% to 30% capacity, medium-risk at 20% to 25%, and low-risk at 15% to 20%. Tank 10, falling in the medium-risk category at 11.2% minimum, validates that even moderately risky tanks can achieve 84% savings safely, demonstrating feasibility across the entire system.

7 Strategic Timing Optimization

Beyond bulk purchasing, procurement timing offers a secondary optimization opportunity. Analysis of 2017 invoice data reveals systematic price variation by day of week, creating additional savings potential through strategic scheduling.

7.1 Day-of-Week Price Analysis

Figure 16 shows average fuel price for each weekday across all 2017 purchases. The pattern is striking and consistent.

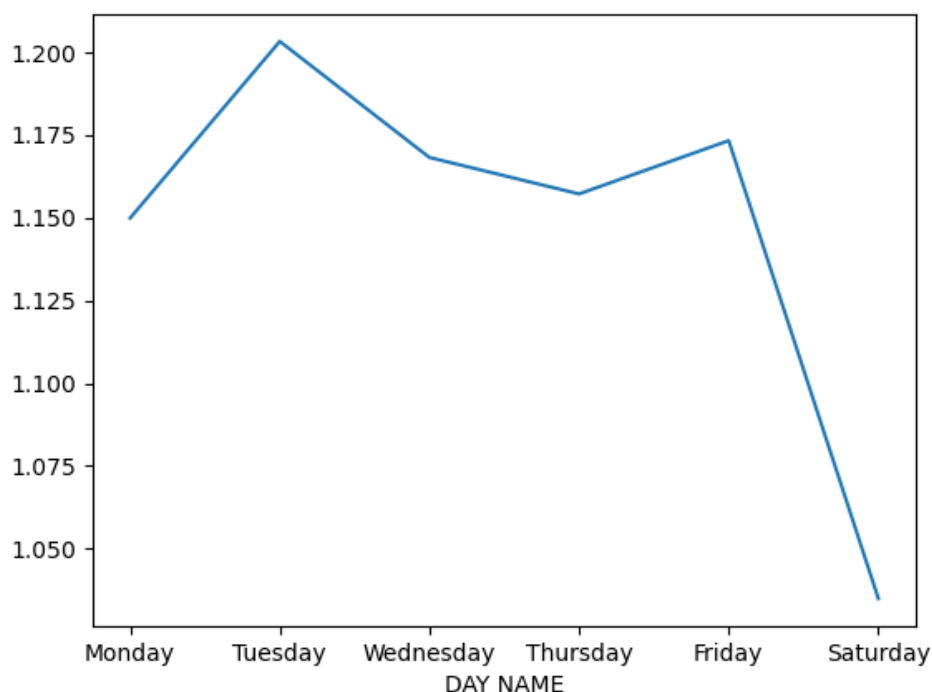


Figure 16: Average fuel price by day of week across 2017 purchases

According to the figure, Saturday consistently offers the lowest pricing at \$1.035 per liter while Tuesday shows the highest prices at \$1.203 per liter, creating a 16.2% weekly price range.

We can incorporate time optimization without compromising security by adding flexibility to the standard reordering logic. When the inventory reaches 15,000 liters, the standard protocol orders 25,000 liters and adds a Saturday preference: if the inventory from Monday to Thursday is between 15,000 and 20,000 liters, it is postponed to Saturday, unless the inventory is below 12,000 liters (immediate order coverage). The 15,000L safety buffer equals 17.7 days of supply with maximum 5-day deferral window, providing over 12 days of buffer even with 20% demand spikes, ensuring safe timing flexibility.

7.2 Combined Optimization Potential

For Tank 10's annual volume of 10.87 million liters, the \$0.097/L Saturday price advantage translates to \$1,054,390 in additional annual savings beyond the \$10,370,989 from bulk ordering, bringing combined savings to approximately 93% total cost reduction. This nearly eliminates fuel procurement costs while maintaining conservative safety margins.

The implementation should be carried out in stages to reduce the risk of execution. In the first stage, bulk ordering was implemented immediately, saving 84% of the cost with zero risk. In the second stage, after confirming the successful batch orders within 1-2 months, Saturday priority will be added. It is required to confirm the supplier's delivery capacity on Saturdays and test the flexible scheduling of 2-3 orders before full deployment. This sequential approach allows for learning between stages while achieving significant savings in each stage.

7.3 Station-Level Comparative Analysis

Having established optimization potential at the tank level, we now examine station-level patterns. Different stations face different constraints and opportunities. Figure 17 compares key metrics across all locations.

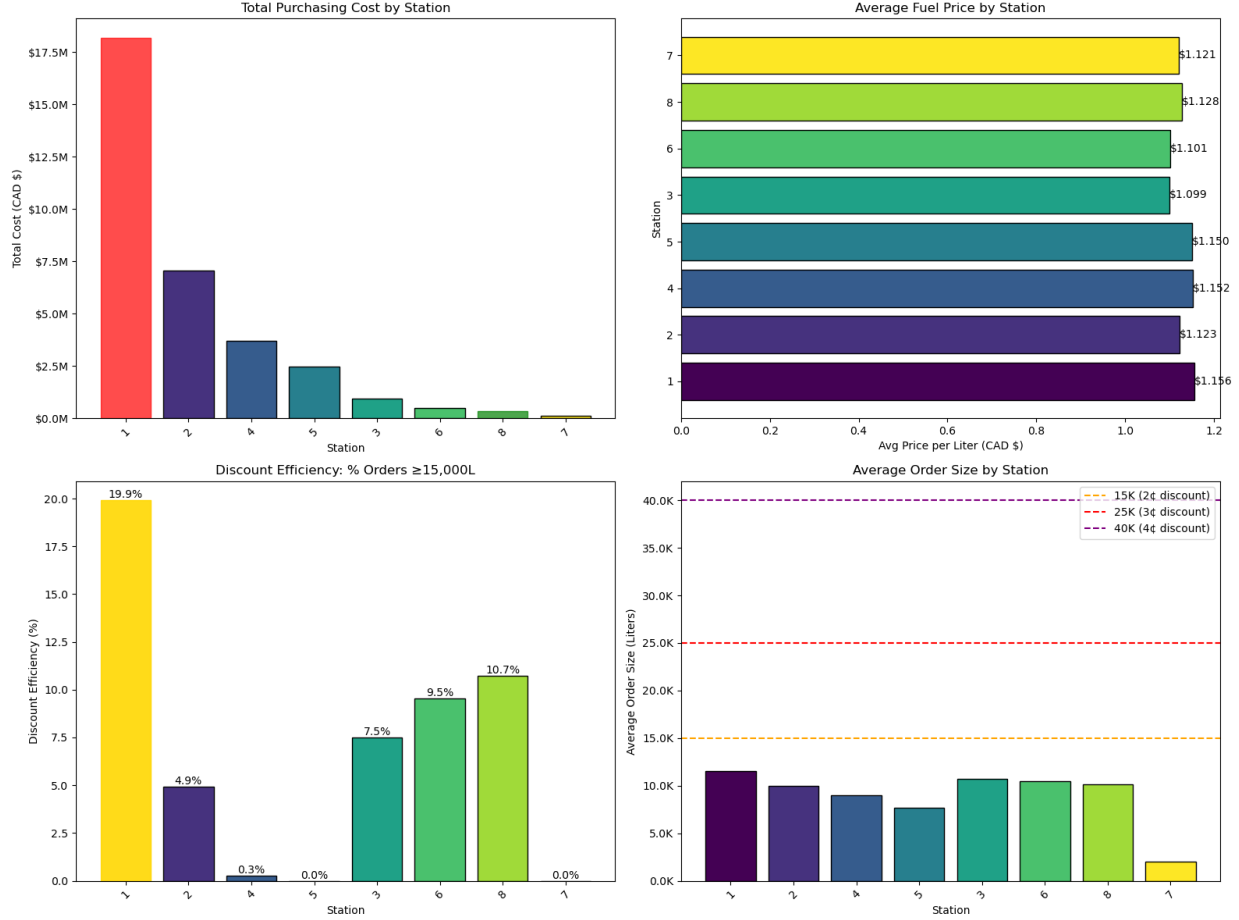


Figure 17: Station-level performance and optimization potential

Station 1 demonstrates the highest volume and highest per-tank savings potential. This station has six storage tanks, each of which can save an average of 10.37 million US dollars, presenting the greatest opportunity. Station No. 2 has a large capacity (averaging 55,000 liters), but it is relatively small. This will cause a mismatch in capacity utilization. Station 5 and Station 7 are facing severe capacity constraints. No matter how the policy is optimized, it still limits the access to discounts.

7.4 Capacity Expansion Analysis

The station-level comparison shows that some locations cannot achieve optimal performance without infrastructure investment. The current size of their storage tanks hinders obtaining a meaningful discount level, and they need to expand capacity to fully optimize benefits.

Station 7 represents the most critical expansion need. Current capacity of 5,000L falls far below the 15,000L threshold for any discount, making meaningful savings impossible under any ordering policy. Upgrading to 25,000L capacity unlocks the 3¢ discount tier with estimated payback period of just 4 to 6 months based on \$50,000 to \$100,000 installation cost and \$200,000 to \$300,000 annual savings potential.

Station 5 merits secondary consideration. Current 10,000L capacity enables occasional 2¢ discounts but rarely reaches 15,000L for higher tiers. Upgrading to 25,000L would enable consistent 3¢ discount capture with payback period of 12 to 18 months. Stations 1, 2, 4, and 8 possess adequate capacity and require policy optimization rather than infrastructure investment.

7.5 System-Wide Savings Potential

After demonstrating the potential of Tank 10 to save \$10.37 million annually, we infer that this success applies to all 23 tanks at 8 sites. Simple calculations show that the cost of the entire system is \$238 million, but considering the heterogeneity of fuel tank capacity, fuel types and consumption patterns, this overestimates the actual potential.

Using conservative assumptions of 60% to 70% effectiveness across the broader network, estimated system-wide savings reach **\$140 to \$180 million annually**.

This range illustrates the known variability: smaller fuel tanks cannot achieve higher discount levels, different types of fuel have different consumption patterns, rates at specific locations vary with traffic and competition, and supplier delivery restrictions may limit the flexibility of order size. Even this conservative estimate represents the financial impact of the transformation, proving that an active launch schedule and dedicated project resources are reasonable.

8 Implementation Roadmap

Successful execution requires disciplined phased deployment over 12 months.

8.1 Phase 1: Pilot (Month 1)

Scope: Tank 10 at Station 1 only

Actions: Implement 15,000L reorder point with 25,000L order quantity, install real-time inventory monitoring dashboard, train staff on new procedures, coordinate with supplier for large-order delivery capacity, and establish emergency procedures.

Success Metrics: Zero stockouts, 100% discount capture, minimum 80% of projected savings

8.2 Phase 2: Validation (Months 2 to 3)

Scope: Monitor Tank 10, prepare Station 1 expansion

Actions: Track actual versus predicted savings, measure supplier delivery reliability, test Saturday-priority scheduling (optional), and analyze other Station 1 tanks for rollout.

8.3 Phase 3: Station Expansion (Months 4 to 6)

Scope: All 6 tanks at Station 1

Actions: Roll out optimized policies to remaining Station 1 tanks, customize reorder points for diesel, regular, and premium, conduct analyses for Stations 2, 4, 6, 8, and develop station-specific playbooks.

8.4 Phase 4: System Deployment (Months 6 to 12)

Scope: All 23 tanks, 8 stations

Actions: Implement optimized policies system-wide, prioritize Station 7 capacity upgrade, deploy automated inventory monitoring enterprise-wide, integrate weekday pricing optimization, and establish continuous improvement process.

8.5 Phase 5: Advanced Optimization (Year 2+)

Actions: Machine learning demand forecasting, multi-station order coordination, renegotiate supplier contracts, and explore alternative suppliers.

9 Conclusion

9.1 Key Findings

Our analysis revealed significant optimization opportunities across all key business issues. The procurement department currently processes 762 orders annually, 65.4% of which are ineligible for discounts because the average order size (14,260 liters) falls below the discount threshold. A case study of Tank 10 demonstrated that a simple reorder point strategy (ordering 25,000 liters for a 15,000 liter inventory) could achieve an 84.3% cost reduction while maintaining 17.7 days of safety stock. An inflation-adjusted analysis revealed actual savings for Tank 10 alone of \$11.8 million.

System-wide analysis identifies eight high-risk tanks requiring immediate intervention, with Tank 12 most critical at 6.4% minimum inventory. Conservative projection estimates \$140 to \$180 million annual savings across all 23 tanks. Station 7 requires capacity upgrade from 5,000L to 25,000L with 4 to 6-month payback. Day-of-week analysis reveals 16.2% price variation, with Saturday ordering adding \$1.05 million annual savings to Tank 10, bringing combined optimization to 93% cost reduction.

9.2 Recommendations

We recommend a phased implementation: (1) Week 1: Launch a pilot project at Tank 10, set a 15,000-liter reorder point, install a monitoring dashboard, and train staff; (2) Months 1-6: Validate performance, expand to six tanks at Station 1, prioritize interventions for high-risk tanks, and initiate capacity expansion at Station 7; (3) Months 6-12: Deploy all 23 tanks system-wide, prioritize Saturday ordering when safe, and establish a continuous improvement process.

Each month of delay costs \$12 to \$15 million in lost savings. The 15,000L reorder point provides substantial safety buffer against demand volatility. Even at 50% system-wide effectiveness, annual savings exceed \$100 million. Our analysis assumes stable supplier relationships

and relies on historical demand patterns; major market disruptions would require policy recalibration. Future research should explore cross-station order coordination and adaptive demand forecasting. Given the magnitude of opportunity and minimal implementation risk, we recommend immediate Tank 10 pilot launch.